LAMMPS official dem10-stack

Preloaded elastic stack | Unequal masses, three contacts | Official tag unvetted | 29 October 2026

PASS	All four runs evaluated with the official final-time 0.05% total mechanical-energy assertion.

Official energy assertion at 20 ms

Run	Final E (microJ)	Final error (%)	Max error (%)	Official
CPU dt	7.615302319	0.0035325	0.0047021	PASS
GPU dt	7.615302319	0.0035325	0.0047021	PASS
CPU dt/2	7.615505900	0.0008593	0.0011755	PASS
GPU dt/2	7.615505900	0.0008593	0.0011755	PASS

E0=7.61557133974e-06 J; abs(E(20ms)/E0-1) <=0.0005. Both spheres' kinetic/gravitational energy and each of the three springs are counted once. This limit applies to total energy, not position/velocity accuracy. Continuous modes quantify accumulated phase error separately.

Official inputs and proper coordinate rotation

Quantity	Unchanged official value / native mapping
Spheres / masses	R=0.5 mm; densities 20000/10000 kg/m^3. Lower/upper mass=1.047197551e-05/5.235987756e-06 kg. Initial velocities and spin zero.
Geometry / rotation	Official centers y=0.45/1.35 mm; fixed planes y=0/1.8 mm. Proper rotation (x,y,z)->(x,-z,y), determinant +1, maps source Y to native Z for positions, forces and gravity.
Contacts / gravity	kn=1000 N/m for floor, sphere pair and roof. e=1 gives automatic zero damping; manual overrides off/zero. Tangential stiffness, friction, resistance and body damping zero; g=9.81 m/s^2.
Time / full outputs	dt=2/1 us to 20 ms; 10000/20000 steps. Every numerical step and t=0: 10001/10001/20001/20001 native frames, total 60004. Four loadable projects and full ZIP retained.

Initial precompression and actual solver bootstrap

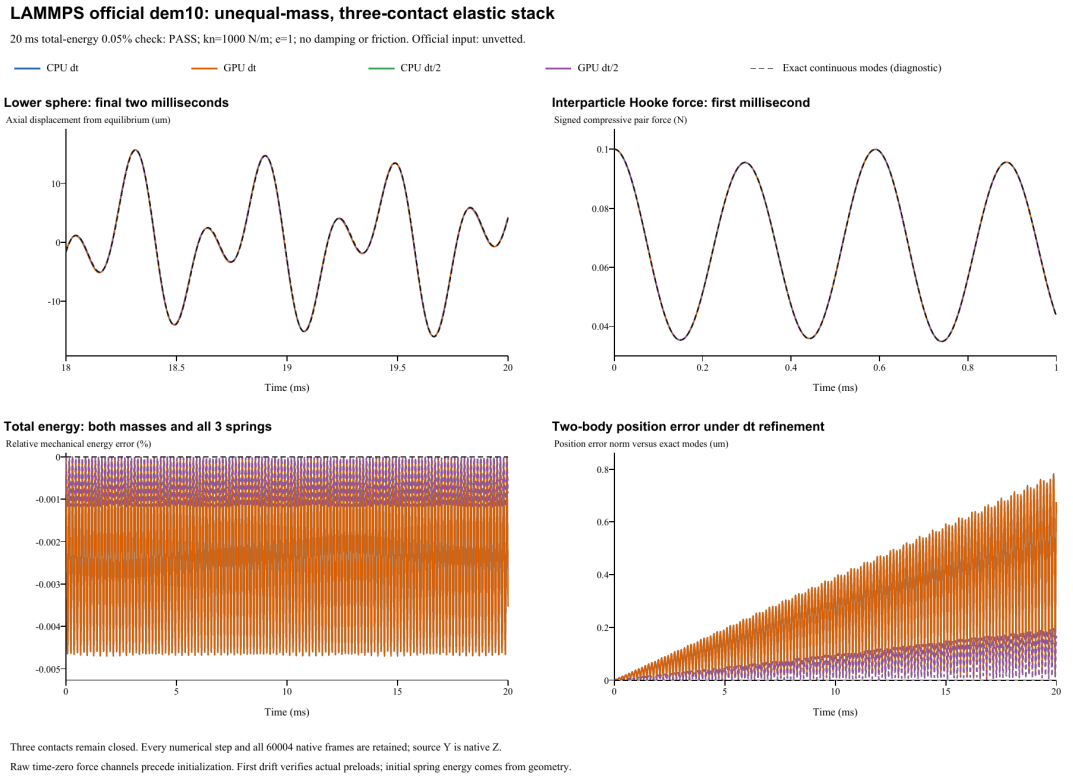
Initial floor/pair/roof overlaps=50/100/50 micrometers; forces=0.05/0.10/0.05 N. Net contact forces: lower -0.05 N, upper +0.05 N; gravity is separate. Initial spring/gravity energies=7.5e-06/1.155713397e-07 J. Unequal initial loads cause undamped oscillation.

Raw t=0 sampling precedes initialization and retains zero force/contact placeholders. Native initialize assembles physical preloads before the first half kick. The first displacement verifies these forces plus gravity. Only derived impulse quadrature uses geometric F0; E0 comes from geometry. Raw data stay unchanged, with no warm-up, force injection or fitted initial conditions.

Reference qualification and source states

Pinned official input | Analytic checker | Source/bootstrap audit.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; YAML blob 405583bd3e27294ab46506f6ba242485a756de33. Official tag: unvetted. LAMMPS was not rerun locally. Five saved checkpoints at 4/8/12/16/20 ms are supplemental; epsilon 1e-10 is same-code precision. This input lies outside the unchanged historical 24-input catalog.

LAMMPS dem10-stack: native histories



Three-contact force and impulse checks

Run	Force residual (N)	Impulse residual	Mode pos. norm (um)	Mode vel. norm (m/s)
CPU dt	1.4e-15	4.2e-20	0.78434	0.016596
GPU dt	1.4e-15	2.8e-20	0.78434	0.016596
CPU dt/2	1.4e-15	2.7e-20	0.19608	0.0041492
GPU dt/2	1.4e-15	5e-20	0.19608	0.0041492

CPU/GPU whole-history maximum position / velocity / contact-force / energy differences are 9.11e-18 m / 1.66e-13 m/s / 9.65e-15 N / 3.83e-19 J.

Halving dt reduces maximum whole-trace relative energy drift by 3.99998; continuous normal-mode position / velocity errors reduce by 4.00009 / 3.99971. These modal comparisons are supplemental and use no fitted phase or coefficients.

Maximum total gravity-and-wall impulse residual is 5.06e-18 kg m/s. The pair contact cancels internally; fixed floor/ceiling loads and gravity supply external impulse. Fixed walls do no work, while particle momentum alone is not conserved.

Fn=kn*d and U=kn*d^2/2 are checked for all three contacts; all overlaps stay positive. Maximum first-drift bootstrap residual is 0 m. Continuous modes quantify transient/phase accuracy without fitting. Raw initial zero force is not substituted for physical startup force.

Complete results and scope

All 60004 native frames, run/project manifests, raw t=0 sample, every-step particle/contact CSV and CPU/CUDA logs are retained. Real SimulationSession uses linked FunDEM 0.4.3 native libraries; installed GUI executable byte identity is not established.

Run manifest | Analysis JSON | Checkpoints | Complete native ZIP | Frame/hash audit

Scope: LAMMPS dem10-stack preloaded elastic contact, unequal masses, correct startup, energy and force integration. Friction, nonzero damping and rotational resistance remain inactive.